

Investor Demand and Dynamic Characteristic Compensations

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Abstract

Standard factor models and theories fail to explain the compensation discrepancy between persistent and transitory components of firm characteristics. We examine which drivers of investors' portfolio allocation choices contribute to the discrepancy by decomposing returns of characteristic-sorted portfolios into different demand-driven components: persistent demand shocks, which reflect idiosyncratic drivers such as emotion and liquidity needs, and demand for current and lagged characteristics, which reflect systematic drivers such as investment mandates and trading costs. We find that persistent demand shocks and demand for characteristics contribute significantly and oppositely to the discrepancy. Further, we provide new evidence that investors have a negative correlation between their demand for current and lagged characteristics.

1 Introduction

Standard factor models assume equal return compensation for each marginal increase in firm characteristics (e.g., Kelly, Pruitt, & Su, 2019). However, recent evidence challenges this assumption, showing that compensation varies between persistent and transitory changes in characteristics (Keloharju, Linnainmaa, & Nyberg, 2021; Yara, Boons, & Tamoni, 2023), which we refer to as the "compensation discrepancy". For example, while compensation for persistent changes is higher for book-to-market and size, transitory changes receive higher compensation for investment and profitability. This discrepancy also leads to different characteristic compensations over various horizons, as transitory components of characteristics fade while persistent components remain over time. Yet, the origins of this discrepancy remain unclear, as standard factor models and theories fail to explain it.

In this paper, we address this gap by examining how investors make portfolio allocation choices and identify which drivers of allocation choices contribute to the discrepancy. We focus on explaining the discrepancy for well-known characteristics, including book-to-market, investment, and profitability. Using a unified framework, we decompose the returns of characteristic-sorted portfolios into demand for current and lagged characteristics (reflecting systematic drivers such as investment mandates and trading costs), persistent demand shocks (reflecting idiosyncratic drivers such as emotion and liquidity needs), and other components.¹ We find that the discrepancy is primarily driven by demand for current and lagged characteristics and persistent demand shocks, indicating the importance of these drivers in understanding the dynamic compensations of characteristics across different horizons.

Why can these drivers contribute to the discrepancy? First, while prior papers have not considered investors' demand for lagged characteristics, it is micro-founded by theories such as investment mandates, portfolio adjustment costs, and slow-moving capital and may significantly drive investors' portfolio allocation choices.² Motivated by these theories, we

¹In the framework, investors demand characteristics because they form their beliefs about expected returns and variance of stocks based on the stock characteristics.

²For investment mandates, see e.g., Hodor and Zapatero (2023). For portfolio adjustment costs, see

analyze investors' demand for lagged book equity, investment, and profitability across one- to four-quarter lags.

Additionally, investors' demand for lagged characteristics may contribute to the compensation discrepancy. For instance, consider a portfolio created by sorting on a characteristic, such as profitability, and buying (selling) stocks with the highest profitability (the lowest profitability) after the sort. At the portfolio level, we find its profitability values increase before the sorting date, peak on the sorting date, and then decline afterward on average. This indicates that the dynamics of current and lagged profitability differ: after the sorting date, current profitability decreases monotonically, while lagged profitability increases for several periods before starting to decline. Assume there are two groups of investors, A and B. Group A positively demands current profitability, while Group B positively demands lagged profitability. After the sorting dates, Group A will gradually reduce its demand for stocks in that portfolio, while Group B will initially increase its demand before eventually decreasing it. Since the groups' demands initially move in opposite directions and then align, the price impact will be smaller at first but larger over longer horizons, contributing to the compensation discrepancy.

Second, extensive behavioral literature suggests that investors' biases can lead to persistent demand shocks (e.g., Barberis, Shleifer, & Vishny, 1998; Hong & Stein, 1999). If the rate of bias correction does not align with the rate of change in the characteristic, this discrepancy will arise. For example, if the correction rate is higher in the early stage, compensations for the characteristic at short horizons will be greater than those at longer horizons.

Third, any non-linearity in the relationship between current characteristics and expected returns can contribute to the compensation discrepancy, as standard factor models assume a linear relationship between expected returns and characteristics. In our framework, we theoretically show that the demand-driven price impact of current characteristics can exhibit

e.g., Bacchetta, Tische, and Van Wincoop (2023), Davis and Azarmsa (2023), and Gârleanu and Pedersen (2013). For slow-moving capital, see e.g., Chaudhry (2023), Duffie (2010), and Mitchell, Pedersen, and Pulvino (2007).

such a non-linear relationship.

To separately examine the impacts of these drivers on compensation discrepancy, we extend the demand system framework in Kojien and Yogo (2019) to first decompose the stock returns into counterfactual price impacts driven by changes in current characteristics, lagged characteristics, latent demand, and others. We then aggregate these counterfactual price impacts at the portfolio levels to decompose the returns of characteristic-sorted portfolios. To test whether any driver contributes to the compensation discrepancy, we follow the approach of Yara et al. (2023), regressing the returns of portfolios created from old sorts on new sorts. A positive alpha suggests higher compensation for the persistent components, while a negative alpha suggests higher compensation for temporary components.

To this point, we have two sets of results. First, regarding which driver contributes to compensation discrepancy, we find that latent demand and lagged characteristics, significantly and oppositely contribute to the compensation discrepancy for book-to-market, profitability, and investment. Additionally, we observe non-linearity in the compensation for current characteristics. In other words, the compensation for a marginal change in current characteristics remains variable, even after controlling for lagged characteristics and latent demand. This suggests that the expected return driven by demand for characteristics is a non-linear function of the current characteristics themselves. Furthermore, we find that other components do not contribute to the compensation discrepancy.

Second, we confirm that lagged characteristics play a significant role in investors' portfolio choices. Notably, there is a significant negative correlation between demand for current characteristics and lagged characteristics across all institutions. This negative correlation is consistent with theories such as Barberis et al. (1998), where investors overreact to growth in fundamentals. Specifically, given current values of characteristics, they perceive higher growth in characteristics when lagged values of characteristics are low.

This paper contributes to three branches of literature. First, we contribute to the evolving demand system asset pricing framework pioneered by Kojien and Yogo (2019), who use

holdings data to jointly explain the behavior of prices, characteristics, and demand. Tamoni, Sokolinski, and Li (2023) estimate a return variance decomposition of characteristic-sorted portfolios to quantify the relative importance of demand-driven components. Chaudhry (2023) examines the gradual adjustment of asset prices in response to changes in growth expectations. It proposes a reduced-form model where investor demand depends on both current and past expected returns, capturing slow-moving capital dynamics and adjustment costs. Huebner (2023) use demand systems to quantify the relative importance of persistent demand shocks and the term structure of demand elasticities on return momentum. Our contribution to this branch is twofold: First, we show new evidence that investors care for lagged characteristics in their portfolio allocation. Additionally, we are the first to examine whether and how different demand-driven components contribute to the dynamic compensation of three important characteristics.

Second, we contribute to the developing literature that seeks to understand the dynamics of characteristic compensation over long horizons. While most papers focus on short-term return predictability of characteristics within twelve months (Chen & Zimmermann, 2021), recent evidence suggests the challenges in understanding return predictability over long horizons. Keloharju et al. (2021) decompose a wide range of characteristics into their persistent and transitory components, arguing that only the average transitory component predicts returns. Liu, Moskowitz, and Stambaugh (2021) further suggest that the mispricing captured by characteristics is more transitory than risk. Baba-Yara, Boons, and Tamoni (2024) argue that the relative compensation for persistent and transitory components varies across characteristics, both in magnitude and sign. We contribute to this literature by showing that three key drivers of investors' allocation choices (demand for current characteristics, lagged characteristics, and latent demand) are crucial in return compensation across different horizons.

Third, we contribute to the literature on characteristics-based factor models, which aim to jointly explain characteristics and stock prices. While most models assume a linear mapping

between expected stock returns and characteristics (e.g., Kelly et al., 2019), others suggest a nonlinear mapping (e.g., Gu, Kelly, & Xiu, 2021). Our contribution to this literature is demonstrating that one source of the nonlinear mapping is the price impact of investor demand for current characteristics as it contributes to the compensation discrepancy.

The remainder of the paper is organized as follows: Section 2 explores the theoretical motivation for examining lagged characteristics and the intuition for why different demand-driven components may contribute to the compensation discrepancy. Section 3 elaborates on data construction. Section 4 elaborates on empirical design. And Section 5 presents the results.

2 Theory and Economic Intuition

This section accomplishes three objectives. In Section 2.1, we provide a brief overview of the theories and economic mechanisms that explain why investors might demand lagged characteristics. We categorize these mechanisms into behavioral biases, market frictions, and investor heterogeneity. Section 2.2 demonstrates how each component may contribute to the compensation discrepancy.

2.1 Motivation for the Demand for Lagged Characteristics

2.1.1 Slow-Moving Capital

Investor demand can be a function of lagged characteristics due to the slow adjustment of capital in response to new information or market opportunities. The concept of slow-moving capital, as discussed in papers by Duffie (2010) and Davis and Azarmsa (2023), highlights how frictions such as search costs, institutional barriers, and capital constraints delay the immediate reallocation of resources. These frictions mean that investors cannot respond instantly to changes in expected returns or market conditions, causing their demand to reflect past characteristics. For example, in times of financial distress, capital constraints can

force investors to delay purchasing undervalued assets, as observed in the 2005 convertible bond market collapse (Duffie (2010)). Similarly, intermediary constraints during market downturns can cause a protracted recovery in asset prices, as capital gradually reenters the market to correct mispricings. This slow adjustment process inherently ties current demand to lagged market conditions, as capital moves incrementally over time.

2.1.2 Portfolio Adjustment Costs and Predictability Decay

Portfolio adjustment costs and the presence of predictability decay provide a strong economic rationale for why investor demand is influenced by lagged characteristics. Gârleanu and Pedersen (2013) and Bacchetta et al. (2023) illustrate that when investors face costs associated with rebalancing their portfolios—such as transaction fees, the risk of price impacts from large trades, and operational frictions—they adjust their positions more slowly. This gradual adjustment process leads to a situation where current demand is based not only on current market conditions but also on past expected returns and portfolio allocations.

Gârleanu and Pedersen (2013) describe how investors aim toward a target portfolio that accounts for both current and future expected returns, but due to adjustment costs, they only partially adjust their holdings. This results in a lagged response in demand, where investor demand at any given time reflects a blend of current and past characteristics, as investors spread out their rebalancing actions over time to minimize costs.

In addition, the joint presence of predictability decay, as shown by Gârleanu and Pedersen (2013) and Jensen, Kelly, Malamud, and Pedersen (2022), further complicates the rebalancing process. When predictors of asset returns decay at different rates, investors optimize their portfolios by weighting these predictors according to their persistence. Transaction costs discourage frequent rebalancing, leading investors to place greater emphasis on characteristics expected to persist. By incorporating multiple lags into their decision-making, investors can minimize trading costs while still capturing returns. This strategic balancing of trading costs with the longevity of return predictors results in demand patterns that are

inherently reflective of past characteristics.

2.1.3 Behavioral Biases and Cognitive Mechanisms

Investor demand may respond to lagged characteristics due to cognitive biases and behavioral patterns that influence decision-making over time. Theories proposed by Daniel, Hirshleifer, and Subrahmanyam (1998) and Barberis et al. (1998) emphasize the role of overconfidence, self-attribution, representativeness, and conservatism in investor behavior. Overconfidence leads investors to overreact to their private information while underreacting to public signals. This bias can cause them to rely on outdated or lagged information, driving demand based on past market conditions. Similarly, conservatism, the tendency to slowly update beliefs in response to new information, further solidifies the reliance on lagged data as investors are reluctant to quickly adjust their expectations. These biases create a feedback loop where investor demand is heavily influenced by prior characteristics, causing persistent over- or underreaction in asset prices.

2.1.4 Heterogeneous Agents and Information Diffusion

The interaction between heterogeneous agents with varying information sets and processing abilities also justifies why investor demand is responsive to lagged characteristics. As modeled by Hong and Stein (1999), the presence of newswatchers and momentum traders, who process information at different rates, leads to a cycle of underreaction, momentum, and overreaction. Newswatchers gradually incorporate information into their decisions, causing delayed responses in demand. Momentum traders, who base their actions on past price movements, further reinforce this lagged response by amplifying price trends. This heterogeneity in information processing speeds and investment strategies means that investors naturally base their demand on lagged market characteristics, as they either act on outdated signals or react to trends established by earlier market participants.

2.1.5 Horizon Heterogeneity and Benchmarking

Investor demand is also responsive to lagged characteristics due to horizon heterogeneity and benchmarking, as discussed by Hodor and Zapatero (2023). In a market composed of investors with varying investment horizons and benchmarks, demand for assets is influenced by these differing temporal preferences. Long-term investors may weigh past characteristics more heavily, as they are less concerned with short-term fluctuations and more focused on long-term trends. Benchmarking further reinforces this behavior, as institutional investors aim to meet specific performance metrics that may be based on historical data. This heterogeneity in investment horizons and objectives causes demand to be distributed across multiple lagged characteristics, as different investors respond to different temporal benchmarks and objectives, creating a complex demand structure that is not solely based on current market conditions.

2.2 Intuition for Contribution to the Discrepancy

Let us begin with standard factor models that assume a linear relationship between characteristics and the expected returns of characteristic-sorted portfolios:

$$E_t(r_{X,t+1}) = X_t\lambda + \beta_{X,M}\lambda_M \quad (1)$$

where E_t is a conditional expectation operator. The conditional expected returns of characteristics-sorted portfolios sorted by characteristic X depend on the compensation λ for X_t and the exposure ($\beta_{X,M}$) to the market factor λ_M . In this factor model, marginal changes in $X_{i,t}$ yield the same compensation λ , which is also the null hypothesis of equal compensation between persistent and transitory components of characteristics in Yara et al. (2023).

Now, let us consider a model that decomposes $X_t\lambda$ into different demand-driven components $c_t^{(k)}$:

Literature	Main Paper	Quote	Sign
Slow Capital	Duffie (2010)	"Partial reversals of these price adjustments will occur over time as the process of intermediation continues to 'digest' the initial shocks"	Positive
Trading Cost	Gârleanu and Pedersen (2013)	"More persistent signals... get more weight in the aim portfolio. Investors should trade more aggressively on persistent signals"	Positive
Overconfidence	Barberis, Shleifer, and Vishny (1998)	"Investors underreact to new information in the short term but overreact in the long term, leading to momentum and reversals"	Negative
Momentum Trading	Hong and Stein (1999)	"Momentum traders' attempts to profit from underreaction lead to an eventual overreaction, as prices overshoot long-run equilibrium values"	Negative
Investor Psychology	Daniel, Hirshleifer, and Subrahmanyam (1998)	"Stock prices overreact to private information signals and underreact to public signals, leading to negative long-lag autocorrelations and excess volatility."	Negative

Table 1: Literature Review on Investor Behavior and Demand

$$E_t(r_{i,t+1}) = \sum_{k=1}^K \frac{\partial E_t(r_{i,t+1})}{\partial c_t^{(k)}} E_t(\Delta c_{t+1}^{(k)}) + \beta_{X,M} \lambda_M \quad (2)$$

where $\frac{\partial E_t(r_{i,t+1})}{\partial c_t^{(k)}}$ represents the expected price impact of marginal changes in component $c_t^{(k)}$ and $E_t(\Delta c_{t+1}^{(k)})$ represents the expected changes in component $c_t^{(k)}$. There are K components. Eq. 2 suggests that if a component does not change on average after portfolio sorting dates, its effect on expected returns will be zero, as $E_t(\Delta c_{t+1}^{(k)}) = 0$. Since there are no economic reasons for components of assets under management, demand for characteristics, and shares outstanding to change after sorting dates, they should not contribute to the compensation discrepancy.

Let us denote $E_t(r_{t+1}^{(k)}) = \frac{\partial E_t(r_{i,t+1})}{\partial c_t^{(k)}} E_t(\Delta c_{t+1}^{(k)})$. Then,

$$\frac{dE_t(r_{t+1}^{(k)})}{dX_t} = \frac{\partial E_t(r_{i,t+1})}{\partial c_t^{(k)} \partial X_t} E_t(\Delta c_{t+1}^{(k)}) + \frac{\partial E_t(r_{i,t+1})}{\partial c_t^{(k)}} \frac{\partial E_t(\Delta c_{t+1}^{(k)})}{\partial X_t} \quad (3)$$

The null hypothesis implies that $\frac{dE_t(r_{t+1}^{(k)})}{dX_t}$ is constant for any component. Rejecting the null hypothesis for a component indicates that the component contributes to the compensation discrepancy. Therefore, we next examine how each component may exhibit a non-constant $\frac{dE_t(r_{t+1}^{(k)})}{dX_t}$.

2.2.1 Latent Demand

In Appendix B, we show that the price impact of marginal changes in latent demand is constant and equals one. Therefore, the only way latent demand could contribute to the compensation discrepancy is through a non-constant $\frac{\partial E_t(\Delta c_{t+1}^{(k)})}{\partial X_t}$. This is when changes in latent demand are not linearly correlated to characteristic X_{it} .

2.2.2 Current Characteristics

There are two reasons why current characteristics might contribute to the compensation discrepancy. First, Appendix B demonstrates that the price impact of marginal changes in current characteristics, or X_t ,³ is a weighted average of investors' demand for characteristic X and the derivative of this price impact with respect to X_t could be non-zero. Thus, the product of the first two terms in Eq. 3, $\frac{\partial E_t(r_{i,t+1})}{\partial c_t^{(k)} \partial X_t} E_t(\Delta c_{t+1}^{(k)})$, could vary as X_t changes over time after the sorting dates.

Furthermore, consider a data-generating process for X_t around portfolio sorting dates:

$$E_t(X_{h,t+1}) = \rho_h X_{h,t} \quad (4)$$

where ρ_h characterizes how X_t evolves over different horizons h . Focus on J periods

³Since other current characteristics should not change on average around sorting dates.

before and after the sorting dates, i.e., $-J < h < J$. As shown in Appendix C, characteristics increase before the sorting date, peak at the sorting date, and decrease after the sorting date. Thus, we assume $\rho_h \in (0, 1)$ when $h \geq 0$ and $\rho_h > 1$ when $h < 0$. Then,

$$\frac{\partial E_t(\Delta c_{t+1}^{(k)})}{\partial X_t} = \frac{\partial(E_t(X_{t+1}) - X_t)}{\partial X_t} = \rho_h - 1 \quad (5)$$

Eq. 5 indicates that when characteristic X_t does not decay linearly after the sorting dates, price impacts due to changes in current characteristics will contribute to the compensation discrepancy, as ρ_h is not constant across horizons h .

2.2.3 Lagged Characteristics

Lagged characteristics can contribute to the discrepancy for three reasons. The first two reasons are analogous to those for current characteristics.

Additionally, note that:

$$\frac{\partial E_t(\Delta c_{t+1}^{(k)})}{\partial X_t} = \frac{\partial(X_t - X_{t-1})}{\partial X_t} = 1 - \frac{1}{\rho_h} \quad (6)$$

Suppose there are four lags. Then $1 - \frac{1}{\rho_h} > 0$ for $h < 4$ and $1 - \frac{1}{\rho_h} < 0$ for $h \geq 4$. This means that even if X_t decays linearly after sorting dates, lagged characteristics can still contribute to the compensation discrepancy because X_t moves in opposite directions before and after sorting dates.

Assuming investors have positive demand for both current and lagged characteristics. Eq. 5 and 6 suggest that the price impacts of current and lagged characteristics are opposite in the first three periods after sorting dates, and align in the same direction after the sorting dates. This is consistent with the example provided in the Introduction.

3 Data

In this section, we elaborate on the data and variable constructions.

3.1 Institutional equity holdings

We obtain quarterly institutional equity holdings of all 13F institutional investors between 1980 and 2017 from the Thomson Reuters Institutional Ownership S34 database. We follow Kojien and Yogo (2019), applying the same filters to the data and the same classification of all 13F investors. Institutional investors are classified into six types: investment advisors, mutual funds, banks, insurance companies, pension funds, and other 13F institutions (e.g., endowments, foundations, and non-financial corporations). We attribute the rest of the holding to the residual investors which we call "households". To calculate the household holding for a given stock, we subtract the total number of shares held by all other six investor groups from the total number of shares outstanding. As a result, the aggregate "household" sector captures both direct holdings by households as well as holdings by non-13F institutions.

3.2 Stock characteristics

Monthly data on stock prices, dividends, returns, and shares outstanding are from the Center for Research in Security Prices (CRSP) database. We restrict the sample to stocks with share codes 10, 11, 12, and 18 that trade on the New York Stock Exchange, American Stock Exchange, and Nasdaq Exchange. Accounting data are from the Compustat database. We apply the same rest of filters as Kojien and Yogo (2019). We include the same set of characteristics from Kojien and Yogo (2019): log market equity, log book equity, dividends to book equity, profitability, investment, and market beta. Since we are interested in the impact of lagged characteristics, for book equity, profitability, and investment, we also calculate their values from one to four quarters ago. Finally, we convert monthly stock characteristics to

quarterly and match the data to holdings data.

4 Empirical Design

This section elaborates on the empirical design. In Section 4.1, we estimate investor demand. Section 4.2 clarifies the decomposition of returns into demand-driven components. Finally, Section 4.3 details how we decompose returns of characteristic-sorted portfolios.

4.1 Demand estimation

In our first step, we estimate each investor’s demand curves building on the demand-system framework of Koijen and Yogo (2019). Demand curves are given by:

$$\log \frac{w_{i,t}(n)}{w_{i,t}(0)} = \gamma_{0,i,t} + \gamma'_{1,i,t} x_t^C(n) + \gamma'_{2,i,t} x_t^L(n) + \epsilon_{i,t}(n) \quad (7)$$

where $w_{i,t}(n)$ denotes the portfolio weight of investor i at date t for stock n . $w_{i,t}(0)$ denotes the portfolio weight on stocks other than ordinary common stocks, which are foreign stocks, real estate investment trusts, or those having missing characteristics or returns. $X_t^C(n)$ denote the vector of most recent stock characteristics. $X_t^L(n)$ denote the vector of all the lagged stock characteristics. $\epsilon_{i,t}(n)$ denotes the latent demand, which is the residual in the demand function.

We estimate the coefficients every quarter and employ the same set of instruments as in Koijen and Yogo (2019). Since many institutions have concentrated holdings, the cross section of their holdings may not be large enough to accurately estimate Eq. 7. Therefore, we estimate coefficients individually for large institutions that have more than 1,000 holdings at a date. For the other institutions, we pool them with similar institutions to estimate their coefficients. We group institutions by their types and assets under management conditional on type. We set the total number of groups within each type to target at least 2,000 strictly

positive holdings per group.⁴

4.2 Decomposition of stock returns

In Koijsen and Yogo (2019), the equilibrium log stock price p_t is a function of the log shares outstanding s_t , the firms' characteristics x_t , the investors' assets under management (AUM) A_t , the investors' coefficients on characteristics γ_t (or demand for characteristics) and the investors-specific latent demand ϵ_t :

$$p_t = g(s_t, A_t, x_t, \gamma_t, \epsilon_t) \quad (8)$$

Define the log returns: $r_{t+1} = p_{t+1} - p_t + v_{t+1}$, where $v_{t+1} = \log(1 + \exp(\frac{d_{t+1}}{p_{t+1}}))$ is the log dividend yield. We consider both current (x_t^C) and lagged (x_t^L) characteristics: $x_t = [x_t^C \ x_t^L]$. The realized log capital gain can be decomposed as:

$$p_{t+1} - p_t = \Delta p_{t+1}^{(x^C)} + \Delta p_{t+1}^{(x^L)} + \Delta p_{t+1}^{(A)} + \Delta p_{t+1}^{(\gamma)} + \Delta p_{t+1}^{(s)} + \Delta p_{t+1}^{(\epsilon)} + \phi_{t+1} \quad (9)$$

where

$$\begin{aligned} \Delta p_{t+1}^{(x^C)} &= g(x_{t+1}^C, x_t^L, A_t, \gamma_t, s_t, \epsilon_t) - p_t \\ \Delta p_{t+1}^{(x^L)} &= g(x_{t+1}^C, x_{t+1}^L, A_t, \gamma_t, s_t, \epsilon_t) - g(x_{t+1}^C, x_t^L, A_t, \gamma_t, s_t, \epsilon_t) \\ \Delta p_{t+1}^{(A)} &= g(x_{t+1}^C, x_{t+1}^L, A_{t+1}, \gamma_t, s_t, \epsilon_t) - g(x_{t+1}^C, x_{t+1}^L, A_t, \gamma_t, s_t, \epsilon_t) \\ \Delta p_{t+1}^{(\gamma, s)} &= g(x_{t+1}^C, x_{t+1}^L, A_{t+1}, \gamma_{t+1}, s_{t+1}, \epsilon_t) - g(x_{t+1}^C, x_{t+1}^L, A_{t+1}, \gamma_t, s_t, \epsilon_t) \\ \Delta p_{t+1}^{(\epsilon)} &= g(x_{t+1}^C, x_{t+1}^L, A_{t+1}, \gamma_{t+1}, s_{t+1}, \epsilon_{t+1}) - g(x_{t+1}^C, x_{t+1}^L, A_{t+1}, \gamma_{t+1}, s_{t+1}, \epsilon_t) \\ \phi_{t+1} &= p_{t+1} - g(x_{t+1}^C, x_{t+1}^L, A_{t+1}, \gamma_{t+1}, s_{t+1}, \epsilon_{t+1}) \end{aligned} \quad (10)$$

⁴Koijsen and Yogo (2019) set the total number of groups within each type to target 2,000 strictly positive holdings per group *on average*. Since we have lagged characteristics, we are concerned about collinearity. To reduce this concern, we require a minimum number of 2,000 positive holdings to increase the sample size in estimation.

Koijen and Yogo (2019) call $\Delta p_{t+1}^{(\epsilon)}$ as the extensive margin of latent demand and ϕ_{t+1} as the intensive margin of latent demand. The extensive margin captures the change in the set of stocks held by investors and the intensive margin captures the changes in portfolio weights within the set of stocks.

After estimating the demand for characteristics γ_t based on quarterly holdings, we estimate the decomposition both at quarterly and monthly frequencies. At the quarterly frequency, we estimate the same decomposition as Eq. 10. At the monthly frequency, we observe updated current (x_t^C) and lagged (x_t^L) characteristics each month. However, we do not observe updated γ_t , A_t , or ϵ_t as holding data updates every quarter. Therefore, we estimate counterfactual price change from changes in current and lagged characteristics and attribute the other changes in returns to changes in the other components in Eq. 8 at the monthly frequency. Studying monthly frequency is because Yara et al. (2023) and the other papers on anomalies focus on monthly returns.

4.3 Decomposition of anomaly returns

We create anomaly portfolios based on book equity,⁵ profit, and investment (we are planning to extend the results to other anomalies as well). For each characteristic, at the end of each quarter, we sort stocks into decile groups based on NYSE breakpoints. After each sort, we construct both value-weighted and equal-weighted long-short spread portfolios by buying the stocks in the highest decile and shorting the stocks in the lowest decile. We track their quarterly returns up to five years to match with the horizons studied in Yara et al. (2023). When a stock delists, we reinvest the amount of money in the stock (net of the delisting return) to the rest of the stocks in the portfolio with value weighting. Finally, we repeat the process for each component (x^C , x^L , A , (γ, s) , ϵ , and ϕ) and for the monthly frequency.

⁵We have estimated demand for current market equity, but do not have estimated demand for lagged market equity due to endogeneity concerns.

5 Results

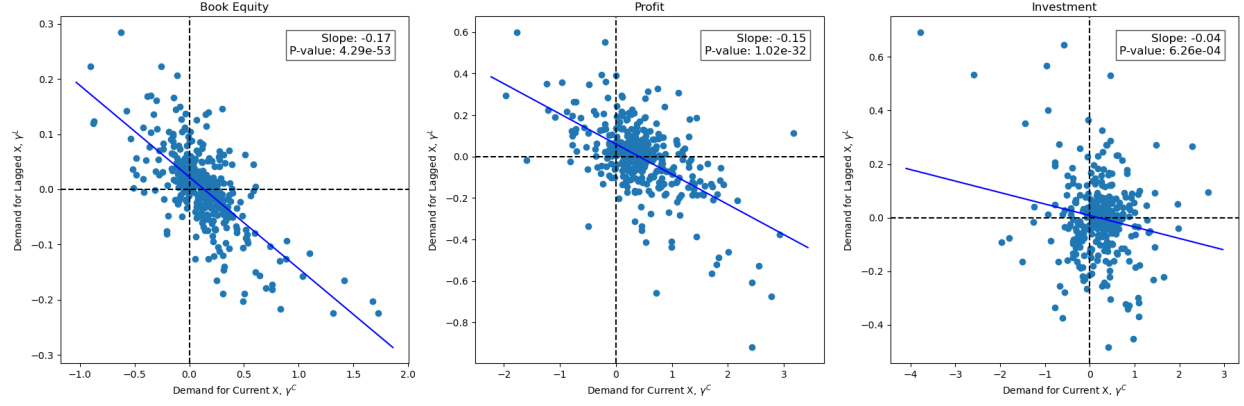
5.1 Demand on current and lagged characteristics

Figure 1 presents the estimated demand for current and lagged characteristics. To simplify interpretations, we average the estimated coefficients for each characteristic across different lags at each date, and then further average these coefficients across all dates for each institution. The distribution of these estimates is plotted for both current and lagged characteristics, with each dot representing an individual institution. We present the estimates for large institutions in Figure (a) and for small institutions in Figure (b).

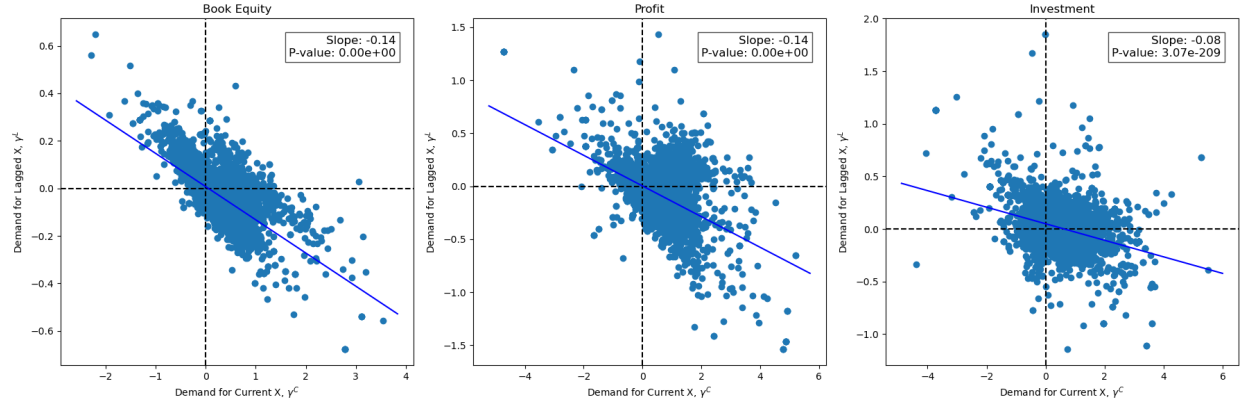
The figures highlight several important findings. First, there is substantial variation in the demand for both current and lagged characteristics across institutions, consistent across all characteristics and for both large and small institutions. This indicates significant heterogeneity in how institutions care about these characteristics.

Additionally, there is a notable negative correlation between the demand for current and lagged characteristics. This indicates that when the current value of a characteristic is higher, institutions tend to reduce their demand if the lagged value of the same characteristic was lower. One potential explanation for this pattern is Barberis et al. (1998), who suggest that investors may overreact to perceived growth in fundamentals. Conditional on the characteristic's current value, a lower lagged value implies greater growth in the characteristic. This prompts investors to increase their holdings, which is reflected in a negative coefficient on lagged characteristics.

The figures also differ across characteristics. For book equity and profitability, the negative correlation is particularly pronounced, indicating that investors are more sensitive to perceived growth or decline in these characteristics. In contrast, the correlation between current and lagged demand for investment is weaker.



(a) Large Institutions



(b) Small Institutions

Figure 1: This figure shows the distribution of estimated demand for current and lagged characteristics γ^C and γ^L , respectively, across institutions for three main characteristics. Figure (a) shows the distribution for large and Figure (b) for small institutions. The plotted dots represent the average coefficients for each institution.

Figure A1 shows the distribution for each lag. The negative correlation between the coefficients on current and lagged characteristics is most pronounced for the first lag. Moreover, there is a positive correlation between the coefficients on current and lagged characteristics for the third lag of book equity.

Figure A2 shows the distribution across different institution types. Results are similar. Results show that there is a significant heterogeneity both across institutions and across characteristics. For example, almost all pension funds have a positive coefficient on current book equity and a negative coefficient on lagged book equity. In contrast, a large proportion of investment advisors have a negative coefficient on current book equity and a positive coefficient on lagged book equity. Additionally, across characteristics, the negative correlation is generally stronger for book equity and profitability but less pronounced for investment. Among institutions, investment advisors and mutual funds exhibit the most robust negative correlation, particularly for book equity and profitability. In contrast, banks, insurance companies, and pension funds typically show positive demand for current characteristics and negative demand for lagged characteristics.

5.2 Which Demand Component Drives the Discrepancy?

In this section, we examine which demand-driven component contributes to the compensation discrepancy and how.

We first verify the results of Yara et al. (2023, BBT) in our quarterly data.⁶ Specifically, we run the following regression:

$$r_{X,(t-20:t-4),t+1} = a_k + b_k r_{X,(t),t+1} + \beta_m r_{m,t+1} + \psi_{X,(t-20:t-4),t+1} \quad (11)$$

⁶BBT have not used quarterly institution holdings data and they examine monthly returns.

where

$$r_{X,(t-20:t-4),t+1} = \frac{1}{5} \sum_{k \in \{4,8,12,16,20\}} r_{X,(t-k),t+1} \quad (12)$$

where $r_{X,(t-20:t-4),t+1}$ represents the equal-weighted returns at $t+1$ of long-short portfolios sorted on characteristic X from 4, 8, 12, 16, and 20 quarters months ago. $r_{m,t+1}$ represents market returns.

In Panel A of Table A1, we show the results are consistent with that of BBT. A positive a_k for book-equity and book-to-market portfolios indicates that the persistent component of those characteristics receives larger compensations than that of the temporary component. A negative a_k for profit and investment indicates the opposite.

We repeat the analysis for each component in Eq. 10. Results from Panel B to Panel G in Table A1 show that a_k is different from zero for current characteristics, lagged characteristics, and latent demand. These non-zero a_k for the components indicate that they contribute to the non-zero a_k observed in the total return, thereby contributing to the compensation discrepancy between persistent and transitory components of characteristics.

Additionally, while the contributions of current and lagged characteristics are in the same direction, they contribute oppositely to latent demand.

Furthermore, as predicted, other components do not have much contribution to compensation discrepancy.

The non-zero a_k may result from each component evolving at a different rate compared to the characteristic used in portfolio sorts. Or, it can come from nonlinear price impacts from the components. To test the latter possibility, we estimate b_k from Eq. 11 using total returns. We then apply the same b_k to different components in Eq. 11. Specifically, we run the following regression:

$$r_{X,(t-20:t-4),t+1} - \hat{b}_k r_{X,(t),t+1} = a_k + \beta_m r_{m,t+1} + \psi_{X,(t-20:t-4),t+1} \quad (13)$$

where $\hat{b}_k$ represents the estimated b_k from total returns.

Table 2 presents the results, showing that current characteristics, lagged characteristics, and latent demand contribute to the compensation discrepancy, also due to non-linear compensations.

Table 2: Decomposition of compensations of characteristics across horizons:

This table presents which demand-driven component contributes to the compensation discrepancy. Panel A shows total returns without decomposition; Panels B to G present results for contemporaneous characteristics, lagged characteristics, AUM, shares outstanding, extensive-margin of latent demand and intensive-margin of latent demand, separately.

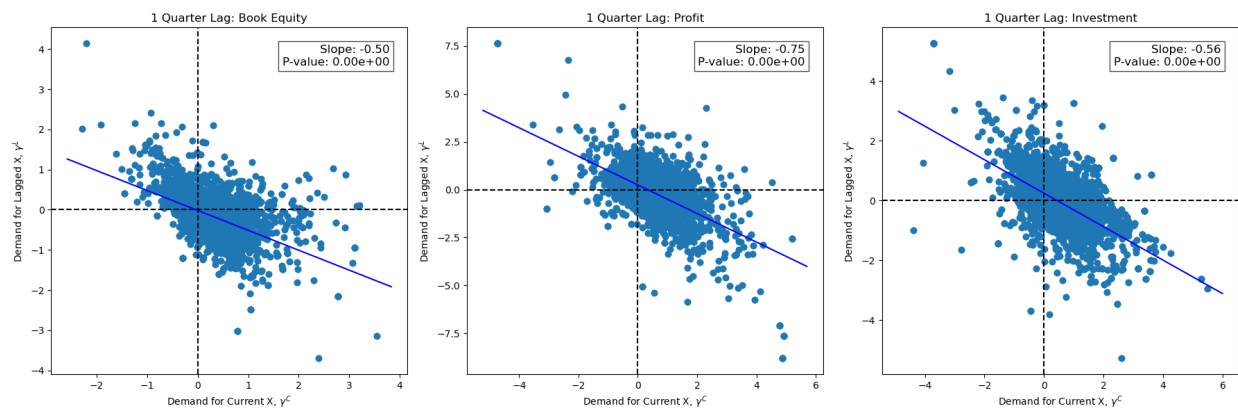
Panel A: Total returns			Panel B: Current characteristics		
Characteristic	a_k	t	Characteristic	a_k	t
Book Equity	0.25*	1.94	Book Equity	0.87***	4.91
Book-to-market	2.00***	5.21	Book-to-market	6.59***	10.59
Profit	-0.41**	(-2.10)	Profit	-2.20***	(-7.52)
Investment	-0.36*	(-1.72)	Investment	0.38*	1.71
Panel C: Lagged characteristics			Panel D: AUM		
Characteristic	a_k	t	Characteristic	a_k	t
Book Equity	-0.21	(-0.70)	Book Equity	0.04	1.2
Book-to-market	1.21***	2.92	Book-to-market	0.05	0.73
Profit	-2.73***	(-6.29)	Profit	-0.04	(-0.62)
Investment	-3.16***	(-11.69)	Investment	0.05	1.08
Panel E: Shares outstanding and coefficients			Panel F: Latent demand: extensive margin		
Characteristic	a_k	t	Characteristic	a_k	t
Book Equity	0.34*	1.66	Book Equity	-0.03	(-0.25)
Book-to-market	-0.4	(-0.83)	Book-to-market	0.92**	2.39
Profit	-0.79*	(-1.86)	Profit	-0.18	(-1.06)
Investment	-0.65	(-1.62)	Investment	-0.07	(-0.28)
Panel G: Latent demand: intensive margin					
Characteristic	a_k	t			
Book Equity	-1.30***	(-3.17)			
Book-to-market	-6.31***	(-6.60)			
Profit	3.14***	4.64			
Investment	3.30***	5.97			

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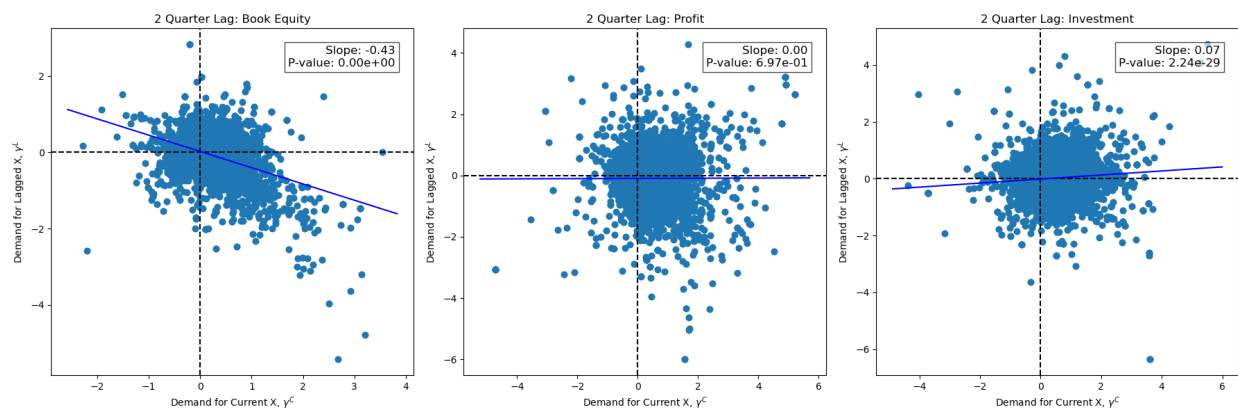
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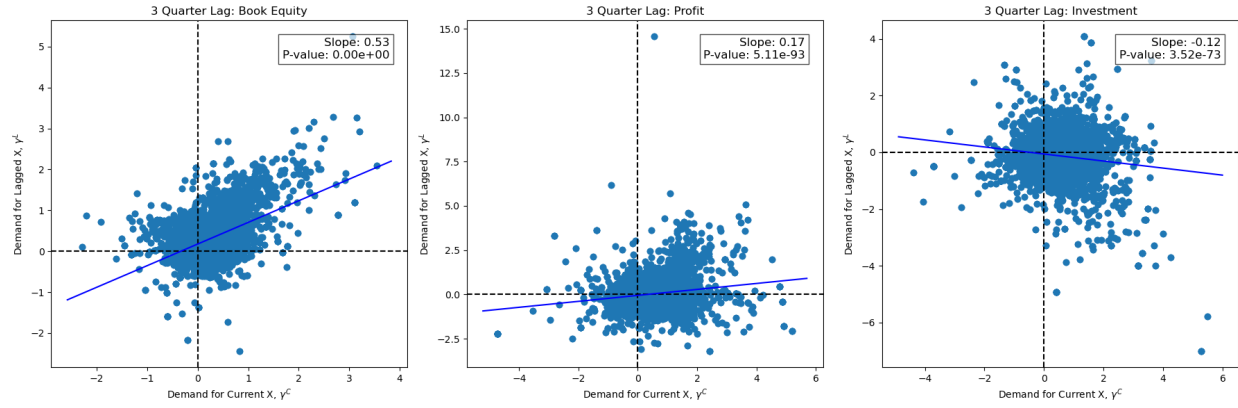
A Appendix: Additional Evidence



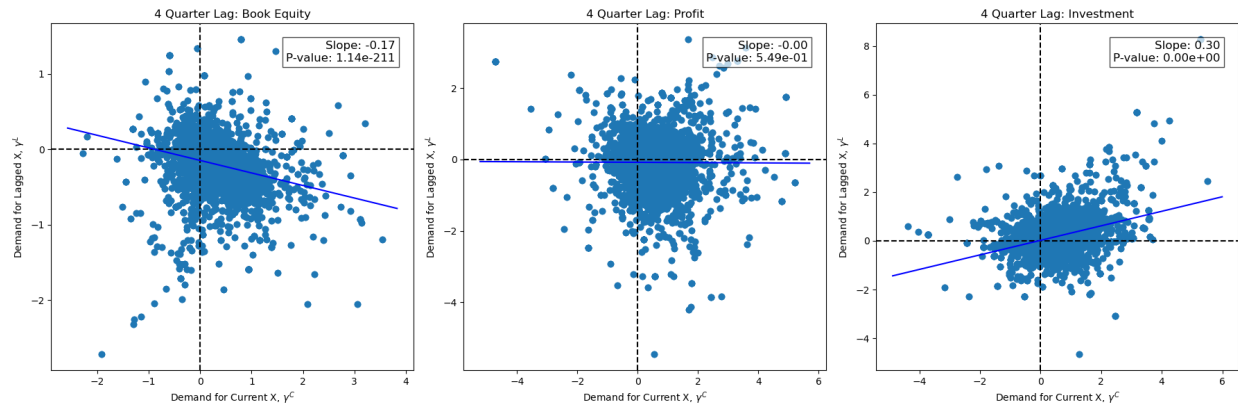
(a) 1 Quarter Lag



(b) 2 Quarters Lag



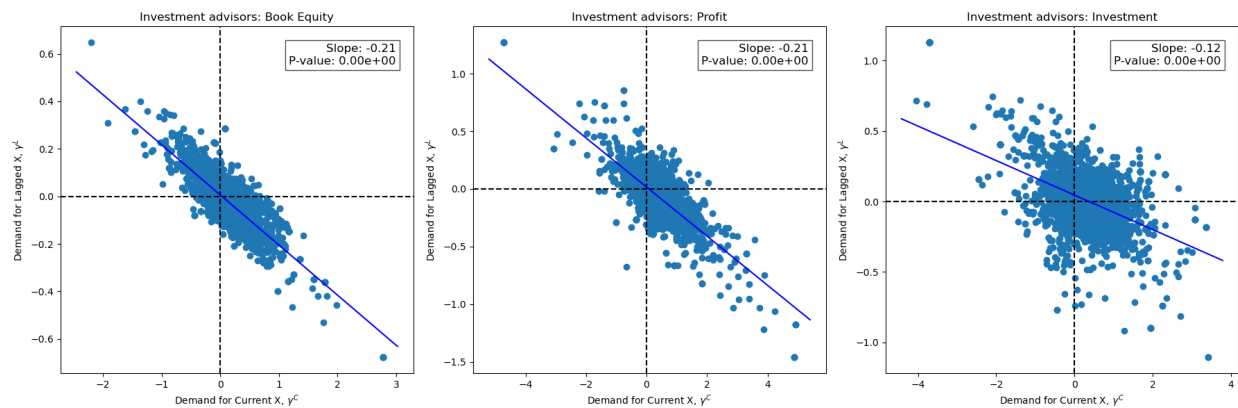
(c) 3 Quarters Lag



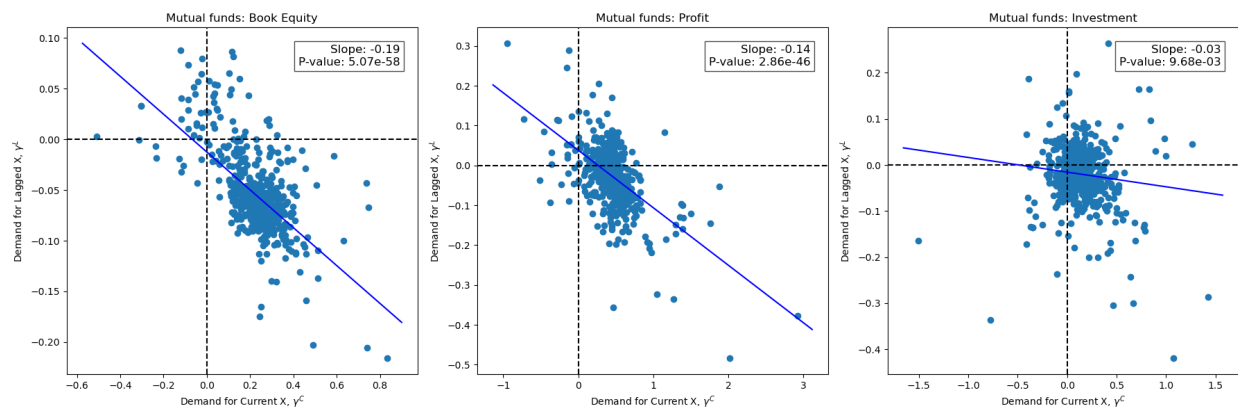
(d) 4 Quarters Lag

Figure A1: Estimates for coefficients γ^C and γ^L with different lags

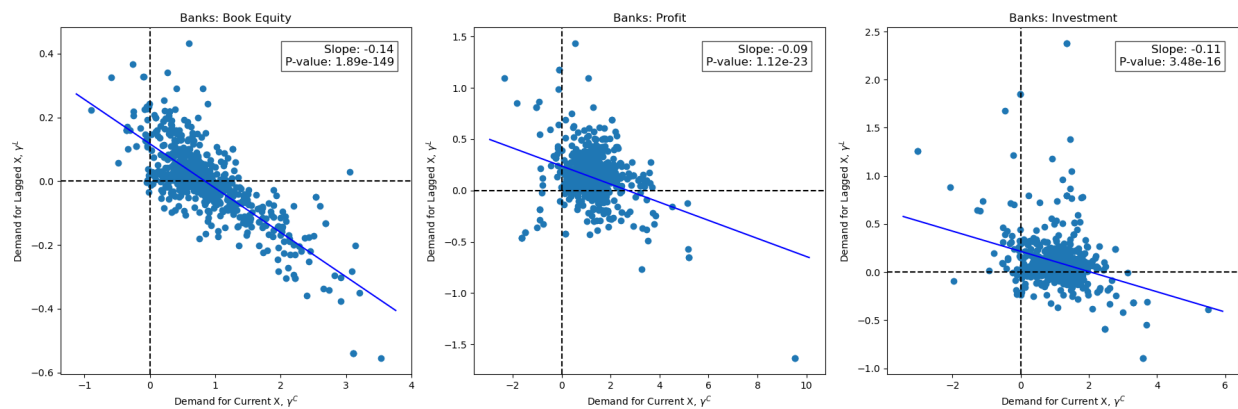
This figure shows the demand coefficients for 4 lags of characteristics. There is a large variation in demand across lags.



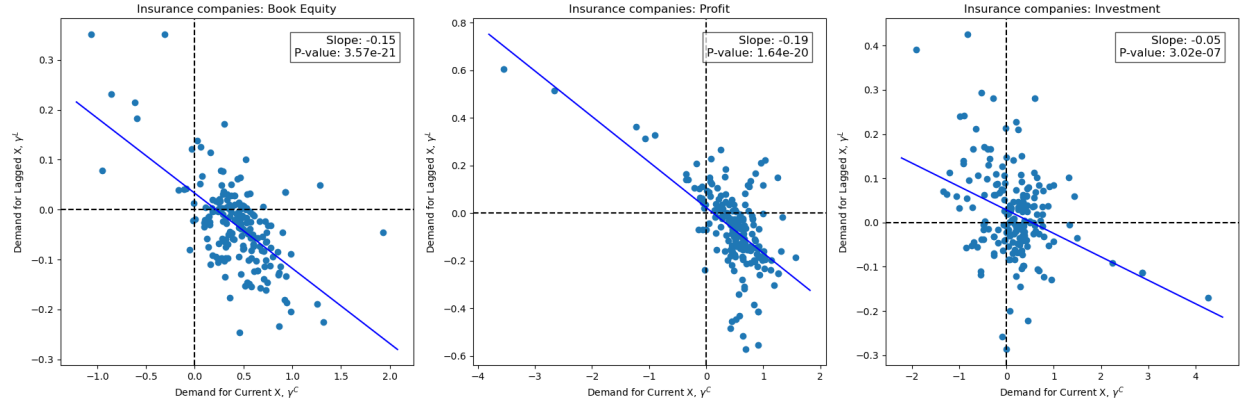
(a) Investment Advisors



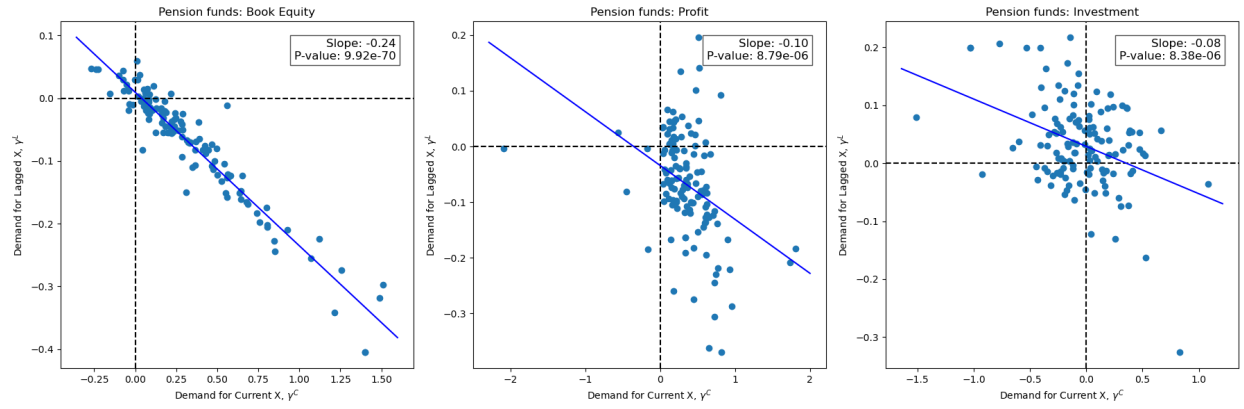
(b) Mutual Funds



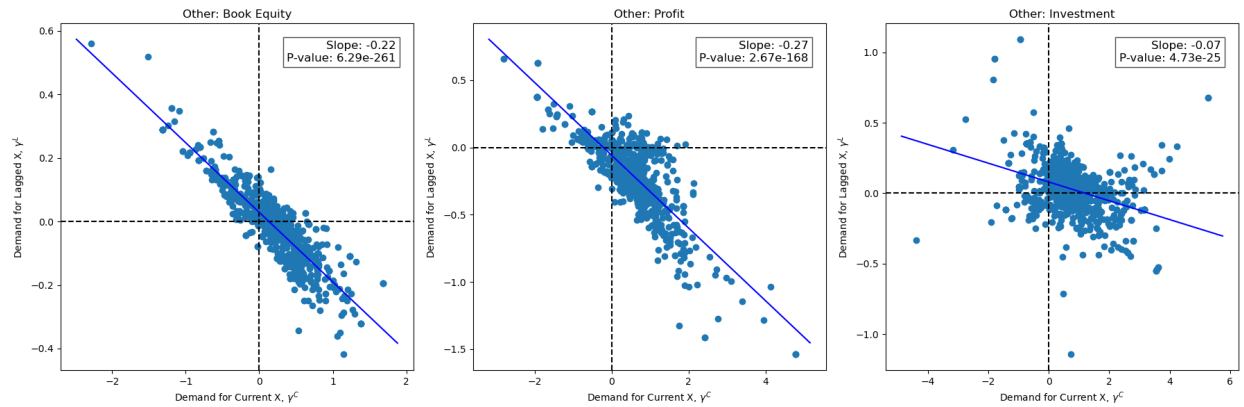
(c) Banks



(d) Insurance Companies



(e) Pension funds



(f) Other Institutions

Figure A2: Estimates for coefficients γ^C and γ^L by investor types

This figure illustrates the distribution of current and lagged characteristics across six types of institutions: investment advisors, mutual funds, banks, insurance companies, pension funds, and other institutions, focusing on three key characteristics.

Table A1: Decomposition of compensations of characteristics across horizons: Restriction on b_k

This table presents which demand-driven component contributes to the compensation discrepancy by restricting b_k as that estimated from regressions for total returns. Panel A shows total returns without decomposition; Panels B to G present results for contemporaneous characteristics, lagged characteristics, AUM, shares outstanding, extensive-margin of latent demand and intensive-margin of latent demand, separately.

Panel A: Total returns			Panel B: Current characteristics		
Characteristic	a_k	t	Characteristic	a_k	t
Book equity	0.25**	2.15	Book equity	0.81***	4.7
Book-to-market	2.00***	5.39	Book-to-market	5.74***	16.36
Profit	-0.41**	(-2.57)	Profit	-1.66***	(-5.55)
Investment	-0.36*	(-1.76)	Investment	-0.37*	(-1.71)
Panel C: Lagged characteristics			Panel D: AUM		
Characteristic	a_k	t	Characteristic	a_k	t
Book equity	0.22	0.89	Book equity	0.07**	2.09
Book-to-market	1.13***	2.66	Book-to-market	0.04	0.58
Profit	-1.35***	(-3.13)	Profit	0.08	1.17
Investment	-3.22***	(-11.49)	Investment	0.10*	1.65
Panel E: Shares outstanding and coefficients			Panel F: Latent demand: extensive margin		
Characteristic	a_k	t	Characteristic	a_k	t
Book equity	0.05	0.24	Book equity	-0.11	(-0.78)
Book-to-market	0.03	0.06	Book-to-market	0.46	1.31
Profit	-0.93**	(-2.19)	Profit	-0.21	(-0.57)
Investment	-0.6	(-1.60)	Investment	0.05	0.15
Panel G: Latent demand: intensive margin					
Characteristic	a_k	t			
Book equity	-0.86***	(-2.82)			
Book-to-market	-5.57***	(-6.81)			
Profit	3.61***	5.71			
Investment	3.69***	7.47			

B Appendix: Proof

This section drives the sensitivity of returns to each demand-driven component.

B.1 Latent Demand

Starting from the logarithmic equation:

$$\ln(P_t(n)) = \ln \left(\sum_{i=1}^I A_{i,t} w_{i,t}(n) \right) - \ln(S_t(n)) \quad (14)$$

$w_{i,t}(n)$ is given by:

$$w_{i,t}(n) = \exp [\beta_{i,t} x_t(n) + \epsilon_{i,t}(n)] w_{i,t}(0)$$

Taking the first derivative with respect to $\epsilon_{i,t}(n)$:

$$\frac{d}{d\epsilon_{i,t}(n)} \ln(P_t(n)) = \frac{\sum_{i=1}^I A_{i,t} w_{i,t}(n)}{\sum_{i=1}^I A_{i,t} w_{i,t}(n)} = 1 \quad (15)$$

Consider asset n as the characteristic-sorted portfolio. This result indicates that a unit change in aggregate latent demand causes a proportional and consistent change in the logarithm of the price of the characteristic-sorted portfolio.

B.2 Characteristics

Taking the derivative of both sides of Eq. 15 with respect to x , we get:

$$\frac{d}{dx} \ln(P_t(n)) = \frac{d}{dx} \left[\ln \left(\sum_{i=1}^I A_{i,t} w_{i,t}(n) \right) - \ln(S_t(n)) \right] \quad (16)$$

Since $w_{i,t}(n)$ is a function of x and the rest are not, the derivative becomes:

$$\frac{d}{dx} \ln(P_t(n)) = \frac{1}{\sum_{i=1}^I A_{i,t} w_{i,t}(n)} \sum_{i=1}^I A_{i,t} \frac{dw_{i,t}(n)}{dx} \quad (17)$$

Taking the derivative of $w_{i,t}(n)$ with respect to $x_t(n)$:

$$\frac{dw_{i,t}(n)}{dx_t(n)} = \beta_{i,t} w_{i,t}(n) \quad (18)$$

Substituting this derivative into equation (17):

$$\frac{d}{dx} \ln(P_t(n)) = \frac{1}{\sum_{i=1}^I A_{i,t} w_{i,t}(n)} \sum_{i=1}^I A_{i,t} \beta_{i,t} w_{i,t}(n) \quad (19)$$

To obtain the second derivative:

$$\frac{d^2}{dx^2} \ln(P_t(n)) = \frac{d}{dx} \left(\frac{1}{\sum_{i=1}^I A_{i,t} w_{i,t}(n)} \sum_{i=1}^I A_{i,t} \beta_{i,t} w_{i,t}(n) \right) \quad (20)$$

Using the product rule:

$$\frac{d^2}{dx^2} \ln(P_t(n)) = -\frac{1}{\left(\sum_{i=1}^I A_{i,t} w_{i,t}(n)\right)^2} \left(\sum_{i=1}^I A_{i,t} \beta_{i,t} w_{i,t}(n)\right)^2 + \frac{1}{\sum_{i=1}^I A_{i,t} w_{i,t}(n)} \sum_{i=1}^I A_{i,t} \beta_{i,t}^2 w_{i,t}(n) \quad (21)$$

The second derivative of $\ln(P_t(n))$ with respect to $x_t(n)$ provides insight into the curvature of the relationship between characteristics and expected returns:

$$\frac{d^2}{dx^2} \ln(P_t(n)) = \frac{1}{\sum_{i=1}^I A_{i,t} w_{i,t}(n)} \sum_{i=1}^I A_{i,t} w_{i,t}(n) \beta_{i,t}^2 - \frac{1}{\left(\sum_{i=1}^I A_{i,t} w_{i,t}(n)\right)^2} \left(\sum_{i=1}^I A_{i,t} w_{i,t}(n) \beta_{i,t}\right)^2 \quad (22)$$

The second derivative is positive when there is significant variance in the $\beta_{i,t}$ coefficients across investors. This suggests that investor demand is highly sensitive to changes in characteristics, resulting in an accelerating (convex) relationship between characteristics and expected returns. The second derivative is negative when the $\beta_{i,t}$ coefficients are relatively uniform across investors. This indicates a concave relationship, where the marginal increase in returns diminishes as characteristics improve. The second derivative is zero when the variance in the $\beta_{i,t}$ coefficients exactly balances the square of their sum, leading to a linear relationship where changes in characteristics result in proportional changes in expected returns.

C Appendix: Characteristic Dynamics Around Sorting Dates

To motivate our analysis of how lagged characteristics may contribute to the compensation discrepancy, we examine how the average characteristic of the stocks in a characteristic-sorted portfolio changes over time. On a given date t , we sort stocks into decile portfolios based on profitability, which we split at NYSE breakpoints. We create a portfolio by buying the stocks in the top decile and shorting the stocks in the bottom decile. For the same set of stocks in this long-short portfolio, we track their average percentile rank from ten years before to ten years after the sorting date. By design, the average rank is nine at the sorting date (ten minus one).

The figure below illustrates that the average percentile rank rises before the sorting date, peaks at the sorting date, and then decays monotonically afterward. Similar patterns are observed for other characteristics. This evidence motivates our subsequent simulation analysis, which assumes that characteristics increase for a few periods and then decrease.

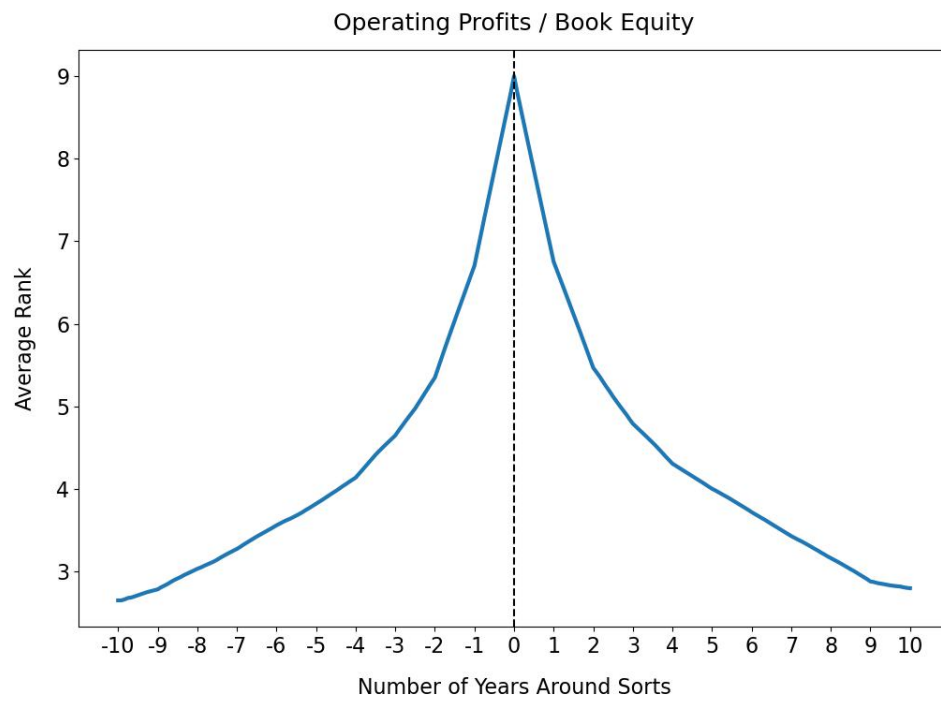


Figure C1: Dynamics in the Average Percentile Rank